Alzheimer's Disease Detection from Brain MRI: A Comparative Study of Five Deep Learning Architectures

Akshaya Bharadhwaj
University of California, Irvine
akshasb1@uci.edu

Leo Zheng
University of California, Irvine
leoz5@uci.edu

Nisarga Patil
University of California, Irvine
nisargap@uci.edu

Tyler Crosbie
University of California, Irvine
tcrosbie@uci.edu

Wonjae Lee
University of California, Irvine
wonjael4@uci.edu

Abstract

Alzheimer's disease (AD) is the leading cause of dementia worldwide, making early and accurate diagnosis critical for timely intervention. In this work, we conduct a comparative evaluation of five deep learning architectures for four-class Alzheimer's stage classification from structural brain MRI using the OASIS-1 dataset: an EfficientNetB0-based CNN baseline, ConvNeXt, Vision Transformer (ViT-B/16), DenseNet-121 with Squeeze-and-Excitation (SE) attention, and an Attention-Gated CNN. Models were trained and evaluated on a shared stratified train/validation/test split, with macro F1-score used as the primary metric to address class imbalance. Grad-CAM analysis was additionally employed to examine the anatomical relevance of learned features.

Among the evaluated models, ViT-B/16 and DenseNet-121 with SE attention were jointly strongest on in-distribution performance (0.342 macro F1; 78.9% and 77.5% accuracy respectively), with the Attention-Gated CNN close behind (0.339). However, zero-shot evaluation on an external MRI dataset revealed substantial domain-shift effects, with most architectures experiencing severe performance degradation and, in some cases, collapsing to single-class predictions. In contrast, the Attention-Gated CNN demonstrated the strongest cross-dataset robustness, maintaining substantially higher performance under distribution shift. These findings suggest that explicit spatial attention mechanisms may contribute more to generalization across MRI acquisition protocols than model scale or in-distribution accuracy alone, highlighting their potential value for clinically deployable AD screening systems.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Transfer learning*.

Keywords

Alzheimer's disease, brain MRI, deep learning, attention gates, Grad-CAM, class imbalance, OASIS

ACM Reference Format:
Akshaya Bharadhwaj, Leo Zheng, Nisarga Patil, Tyler Crosbie, and Wonjae Lee. 2026. Alzheimer's Disease Detection from Brain MRI: A Comparative Study of Five Deep Learning Architectures. In *Proceedings of Neural Networks and Deep Learning (CS 274P).* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

CS 274P, UC Irvine
2026. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Alzheimer's disease is a progressive neurodegenerative disorder and the most common form of dementia, accounting for approximately 60–70% of all dementia cases globally. The World Health Organization estimates that 55 million people currently live with dementia, with nearly 10 million new cases each year. Despite decades of research, no disease-modifying treatment exists, and clinical management depends critically on early detection. Studies show that identifying pre-clinical or early-stage Alzheimer's and beginning supportive intervention can delay functional decline by several years. Traditional diagnostic workflows rely on expensive neuroimaging modalities such as amyloid PET scans, or on invasive cerebrospinal fluid (CSF) biomarker assays, neither of which is practical for widespread screening. Structural magnetic resonance imaging (MRI), by contrast, is non-invasive, widely available, and affordable. It captures the hallmark structural changes of Alzheimer's progression which includes hippocampal atrophy, ventricular enlargement, and cortical thinning, that begin years before clinical symptom onset. However, interpreting subtle structural changes in early disease stages requires specialist expertise, creating a significant bottleneck in clinical workflows globally. Deep learning has demonstrated strong performance on medical image classification tasks, and there is growing interest in applying it to Alzheimer's detection from structural MRI. Yet much of the existing literature evaluates only one or two architectures in isolation, making it difficult to understand the relative strengths of different approaches. Furthermore, clinical deployment of black-box classifiers is increasingly untenable as regulators, clinicians, and patients require explanations for AI-driven decisions, especially in high-stakes settings. This work addresses both gaps through a controlled comparative study. We train and evaluate five architectures on the same dataset, split, and evaluation protocol, enabling direct comparison. We additionally apply Grad-CAM interpretability analysis to the CNN-based models to assess whether each architecture learns clinically meaningful spatial features, and compare this with the explicit attention maps produced by our novel Attention-Gated CNN. Our specific contributions are:

- A fair, controlled comparison of five deep learning architectures spanning CNNs, residual networks, transformers, densely connected networks, and attention-gated models on 4-class Alzheimer's staging from 2D MRI slices
- Rigorous handling of dataset class imbalance through weighted sampling and weighted loss, with evaluation via macro F1-score rather than raw accuracy

- A Grad-CAM interpretability analysis across the CNN-based architectures (Custom CNN, DenseNet-121+SE, and the Attention-Gated CNN), asking whether different inductive biases lead to different spatial attention patterns on brain MRI

2 Data

2.1 The OASIS-1 Dataset

We use the preprocessed OASIS-1 dataset [7] available on Kaggle (`ninadaithal/imagesoasis`). The Open Access Series of Imaging Studies (OASIS) was developed at Washington University School of Medicine and comprises 416 subjects aged 18–96, with T1-weighted 3D structural MRI acquisitions. Subjects include both cognitively normal individuals and Alzheimer's patients at various stages. The dataset is accompanied by Clinical Dementia Rating (CDR) scores, providing ground truth labels for our four-class staging task.

The Kaggle version provides 2D axial MRI slices as JPEG images organized by class, enabling standard image classification pipelines without requiring DICOM or NIfTI processing. The four classes correspond to CDR scores: Non-Demented (CDR 0), Very Mild Demented (CDR 0.5), Mild Demented (CDR 1), and Moderate Demented (CDR 2).

2.2 Class Distribution and Imbalance

The dataset exhibits severe class imbalance, summarized in Table 1. Moderate Dementia, representing the most advanced stage in this dataset, comprises less than 1% of all samples. This extreme imbalance poses a fundamental challenge: a naive classifier that predicts Non Demented for every sample achieves 77.5% accuracy while completely failing the clinically critical rare classes. All models in this study must explicitly address this imbalance.

Table 1: Class distribution and computed loss weights. Weight $w_c = N/(C \times n_c)$, where N is total images, $C = 4$ classes, and n_c is class count.

Class	Images	% of Total	Loss Weight
Non Demented	67,222	77.8%	0.321×
Very Mild Demented	13,725	15.9%	1.575×
Mild Demented	5,002	5.8%	4.321×
Moderate Demented	488	0.6%	44.23×

All models in this study explicitly address this imbalance, though through different mechanisms depending on framework. The PyTorch-based models (ConvNeXt, ViT-B/16, DenseNet-121+SE, and the Attention-Gated CNN) use **WeightedRandomSampler** during training, which oversamples minority classes so each batch is approximately class-balanced, combined with **Weighted CrossEntropyLoss** using the per-class weights in Table 1. The Keras-based Custom CNN instead applies these same inverse-frequency weights via the `class_weight` argument to categorical cross-entropy loss (Section 3.1.2), which penalizes minority-class errors equivalently without altering the sampling distribution. All weights are computed once from `preprocessing.py` and shared across the team.

2.3 Data Split and Data Leakage Prevention

An important methodological consideration for the OASIS dataset is subject-level splitting. The dataset provides approximately 249 slices per subject, and naive random splitting at the image level places slices from the same patient in both training and test sets. Because adjacent slices share near-identical anatomy, the model learns to recognize patients rather than disease stage, inflating test accuracy to 99.97% in preliminary experiments.

We therefore split at the subject level. All slices from a given patient are assigned entirely to one partition. Using `random_state=42`, we perform a stratified 70/15/15 split across 347 subjects (242 train / 52 val / 53 test). The second split (val/test) is not stratified because Moderate Dementia has only 2 subjects total, leaving fewer than 2 per class in the temporary pool. Split indices are saved to `split_indices.json` and shared across all five models.

Additionally, since OASIS slices span multiple anatomical depths and orientations, we filter to axial slices in the mid-brain z-range (slice indices 60–120), retaining slices at the ventricular and periventricular level most relevant to Alzheimer's structural changes. This reduces the dataset to approximately 20,000 training images.

2.4 Preprocessing and Augmentation

All images are resized to 224×224 pixels and normalized using ImageNet statistics (mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$). Training images receive random horizontal flip ($p = 0.5$), random rotation ($\pm 15°$), color jitter (brightness ± 0.2, contrast ± 0.2), and random affine translation ($\pm 5\%$). Validation and test images receive only resize and normalization.

3 Methods: Architecture Design and Training

We train and evaluate five deep learning architectures. Four use ImageNet-pretrained weights with fine-tuning; the custom CNN is to be used as a baseline measuring the value of transfer learning. All models share an identical data pipeline, split, loss function weighting, and evaluation protocol.

3.1 Custom CNN

The custom CNN is implemented using an ImageNet-pretrained EfficientNetB0 backbone and serves as a lightweight baseline for comparison against the larger architectures evaluated in this study. This serves as the fundamental baseline: any model that does not outperform it fails to justify its added complexity, and its performance relative to the four pretrained models directly quantifies the benefit of ImageNet transfer learning on this medical imaging task.

3.1.1 Architecture. Input MRI slices were resized to $224 \times 224 \times 3$ and passed through light image augmentation consisting of small random rotations, zoom, and contrast adjustment. The EfficientNetB0 backbone was loaded without its original classification head and initialized with ImageNet weights. During initial training, the backbone was frozen so that the model learned only the new task-specific classification layers.

The EfficientNetB0 feature extractor outputs a $7 \times 7 \times 1280$ spatial feature map. The custom convolutional layer was appended with 64 filters, a 3×3 kernel, same padding, and ReLU activation. This helps adapt the pretrained image features to the MRI classification task

while preserving spatial feature maps for Grad-CAM visualization. The output is then passed through global average pooling, reducing the spatial feature maps to a compact feature vector and limiting the number of trainable parameters. A dropout layer with rate 0.35 is applied for regularization, followed by a dense softmax classification head with four outputs corresponding to the four dementia conditions. Grad-CAM is computed on the final custom convolutional layer (`gradcam_conv`).

3.1.2 Training Strategy. The shared team preprocessing pipeline was used, including the 70/15/15 split, axial slice filtering over slice indices 60–120, ImageNet normalization, and the class ordering. To handle class imbalance, the model used loss weights returned by `preprocessing.py`, passed into Keras through the `class_weight` argument during training.

The model was trained with the Adam optimizer using an initial learning rate of 10^{-3}, batch size 32, and categorical cross-entropy loss. We trained for up to 15 epochs with early stopping on validation loss and `ReduceLROnPlateau` scheduling to lower the learning rate when validation loss stopped improving. The best checkpoint was selected by validation loss and evaluated once on the OASIS test set.

3.1.3 Interpretability. For interpretability, Grad-CAM was applied to the final custom convolutional layer. This produces class-discriminative heatmaps showing which spatial regions of the MRI slice contributed most strongly to the model's prediction. Due to the model using a convolutional feature map before global average pooling, Grad-CAM can highlight whether the model focuses on clinically meaningful structures such as cortical or ventricular regions rather than irrelevant background or skull artifacts.

3.2 ConvNeXt

ConvNeXt [5] is one of the most widely used baselines in medical imaging research, providing the most direct comparison with published Alzheimer's detection literature. Its residual connections allow effective gradient propagation through layers, enabling deep feature hierarchies from MRI texture and structure. We replace the original 1000-class output head with a $768 \rightarrow 4$ linear classifier and fine-tune using AdamW.

3.2.1 Architecture. The input image of size 224×224 is replicated across three channels to enable the use of ImageNet's pretrained weights in our model without any architectural changes. This is processed by a 4×4 patchify stem convolution that downsamples to 56×56. This mirrors the maxpooling steps in traditional CNNs. Then it is followed by four stages of ConvNeXt blocks. At each stage, the spatial size is halved while the number of channels is doubled. This results in a 7×7 map with 768 channels. We reach the global average pooling layer, which collapses the feature map to a 768 dimensional vector. Finally, this is passed into a custom head comprised of LayerNorm, dropout of 0.35, and a linear layer to our four classes.

3.2.2 Design Modernization. The ConvNeXt architecture is derived from ResNet, which remains one of the most widely used backbones in medical image analysis. ResNet utilizes the residual skip connection, where the block output is $\mathbf{y} = \mathbf{F}(\mathbf{x}) + \mathbf{x}$, given block transformation $\mathbf{F}$ and input vector $\mathbf{x}$. This shortcut allows uninterrupted gradient flow, enabling the training of much deeper networks. The standard ResNet block employs a compression then expansion sequence of three convolutions. First is a 1×1 convolution to reduce channel width by a factor of four. Then comes a 3×3 convolution for spatial feature extraction. Finally, a 1×1 convolution restores the channel width. As seen in Figure 1, each convolution is followed by BatchNorm and ReLU.

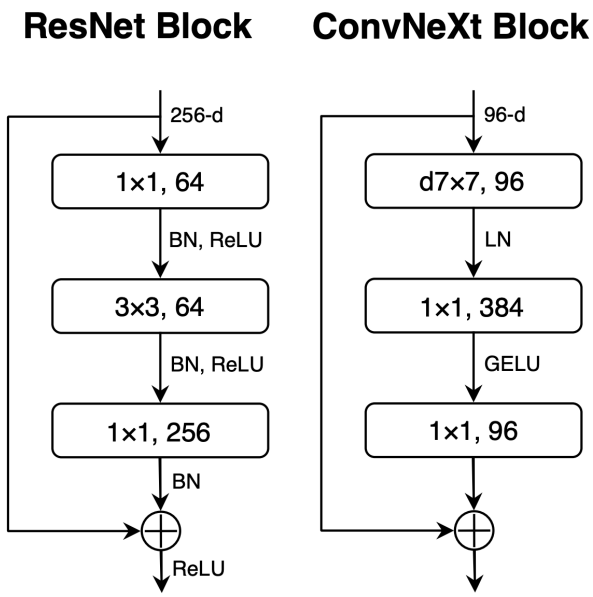


Figure 1: ResNet block vs ConvNeXt block. ResNet applies a 1×1 compress, 3×3 standard convolution, 1×1 expand sequence with BatchNorm and ReLU. ConvNeXt replaces this with a 7×7 depthwise convolution for wide spatial mixing, LayerNorm for per-sample normalization, an inverted 1×1 expand ($\times 4$ channels) followed by a GELU activation and a 1×1 project, with stochastic depth on the residual path. The channel expansion order is inverted relative to ResNet: ConvNeXt expands before projecting rather than compressing before expanding.

While ConvNeXt preserves ResNet's four stage structure and residual connections, it modernizes the design within the block. The authors of A ConvNet for the 2020s [5] identified a system of changes including kernel sizes, inverted channel expansion, modified normalization, and an alternative activation function that aimed to close the performance gap between CNNs and Vision Transformers. With these changes, ConvNeXt attempts to achieve the same level of representation quality as ViTs while removing the need for attention mechanisms and positional encodings. In comparison to ViTs global context through self attention, ConvNeXt gains a wide receptive field through its 7×7 depthwise convolutions and the spatial compression throughout four stages of the blocks. Like ViT, ConvNeXt replaces ResNet's usage of BatchNorm with LayerNorm as it has a more stable normalization under class imbalance.

ConvNeXt also goes for the GELU activation over ReLU for its smoother gradients. By implementing these more modernized techniques, ConvNeXt seeks to replicate the performance capabilities of ViTs while maintaining the foundational of CNNs.

3.2.3 Training Strategy. The parameters were fine tuned from ImageNet weights without any freeze/unfreeze schedule. We used AdamW with a learning rate of $\text{lr} = 10^{-4}$, weight decay of 0.05, accompanied by cosine annealing to regulate model convergence. The shifting of the pretrained weights per step is small enough such that the ImageNet features are mostly retained while adapting to the MRI image domain. In order to mitigate class imbalance, we applied a cross entropy loss weighted inversely proportional to class frequencies as mentioned previously. This allows for the more rare class, Moderate, to carry a much larger weight than that of Non-Demented, so errors cost proportionally more. Additionally, augmentation was performed on the classes such as flips, rotation, and translation to increase the representation of minority classes. Using macro F1 as our metric instead of purely focusing on accuracy enables equality across all four of the classes.

3.2.4 Interpretability. The Grad-CAM visualization was applied to the final ConvNeXt block of stage 4, which captures the final spatial feature maps prior to global average pooling. To isolate the features that the model focused on, gradients of the predicted class score were backpropagated with respect to these feature maps. Global average pooling was then applied to the resulting gradients to determine the weighed importance of each channel. Using this strategy, we are able to visualize the spatial regions that the model concentrated on when making a prediction.

3.3 Vision Transformer (ViT-B/16)

ViT-B/16 [2] processes the MRI image as a sequence of 196 non-overlapping 16×16 pixel patches. Multi-head self-attention across all patches enables global spatial reasoning from the first encoder layer—a fundamental difference from CNNs, which build global context only in deep layers. For Alzheimer's detection, this is particularly relevant because diffuse cortical thinning and ventricular enlargement are whole-brain phenomena that may be better captured by global attention than local convolutional filters.

This model is built on the Vision Transformer (ViT-B/16) [2], initialized from ImageNet-pretrained weights via the `google/vit-base-patch16-224` checkpoint (Figure 2). Unlike convolutional encoders, which aggregate information through local receptive fields, the ViT applies global self-attention from the first layer, allowing any patch to attend to any other. This is well-suited to Alzheimer's MRI, where the discriminative signal is spatially distributed—ventricular enlargement, periventricular change, and cortical thinning—rather than confined to a single region. Because Vision Transformers lack the spatial inductive biases of CNNs and therefore rely on large-scale pretraining to generalize from limited data [2], we fine-tuned from ImageNet weights rather than training from scratch.

3.3.1 Self-Attention Mechanism. Each encoder layer computes scaled dot-product self-attention over the patch tokens. Given query, key, and value projections Q, K, V of the token embeddings:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{1}$$

where d_k is the key dimension [11]. This is computed in parallel across 12 heads whose outputs are concatenated and projected, letting the model integrate evidence from anatomically distant regions within a single layer.

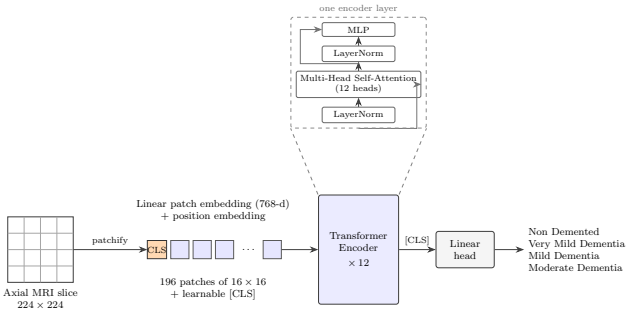


Figure 2: ViT-B/16 architecture. The 224×224 input is split into 196 non-overlapping 16×16 patches, linearly embedded to 768 dimensions, and prepended with a learnable [CLS] token. Twelve Transformer encoder layers (12 heads each) process the sequence; the final [CLS] representation is passed to a linear head reinitialized for the four diagnostic classes.

3.3.2 Architecture. The input is divided into 196 non-overlapping 16×16 patches, each linearly projected to a 768-dimensional embedding and augmented with a learnable [CLS] token and additive position embeddings. The sequence is processed by 12 Transformer encoder layers (multi-head self-attention with 12 heads plus a position-wise feed-forward block), totaling ≈ 86M parameters. The original 1000-way classification head was discarded and replaced with a linear classifier mapping the final [CLS] embedding to the four diagnostic classes, with the mismatched head reinitialized at load time.

3.3.3 Training Strategy. All parameters were fine-tuned end-to-end; no layers were frozen. To address class imbalance we optimized a class-weighted cross-entropy loss with inverse-frequency weights, computed once over the training split and shared across the team's models. We used AdamW [6] with a learning rate of 2×10^{-5}, weight decay 0.01, and a linear schedule with warmup over the first 10% of steps followed by linear decay, a batch size of 32, and mixed-precision (FP16) training for up to five epochs on a single NVIDIA A100 GPU. Early stopping with patience three monitored the validation macro-F1, and the highest-scoring checkpoint was retained.

3.3.4 Interpretability. The convolutional models in this study are interpreted with Grad-CAM, which backpropagates the target logit to the final spatial feature map to produce a class-discriminative heatmap. This does not transfer to the ViT: the encoder operates on a sequence of patch tokens and retains no spatial convolutional feature map for Grad-CAM to weight, so the method is not applicable here. The appropriate analogue for transformer encoders is

an attention-based attribution such as attention rollout [1], which traces patch-to-[CLS] attention across layers; we did not perform this analysis and leave a spatial relevance study of the ViT to future work.

3.4 DenseNet-121 with Squeeze-and-Excitation

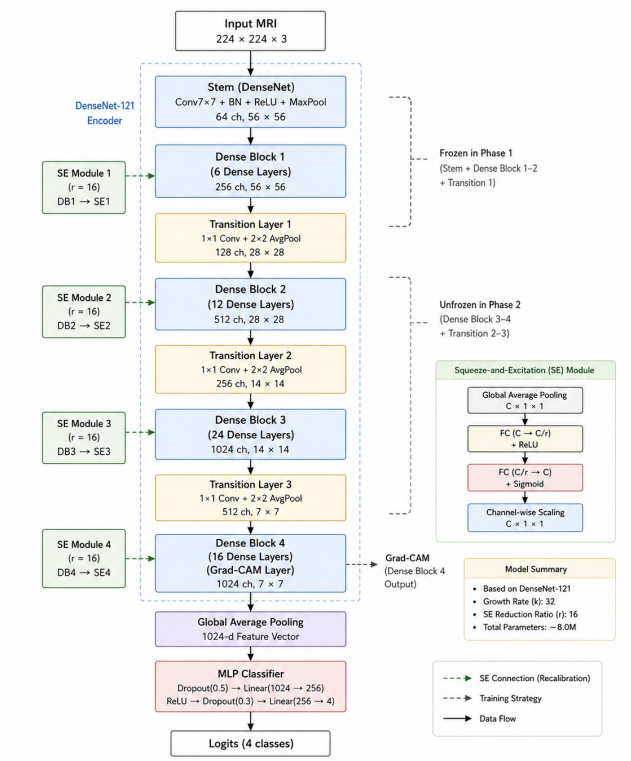


Figure 3: DenseNet-121 + SE architecture for 4-class Alzheimer's staging. A $224 \times 224 \times 3$ axial MRI slice passes through an ImageNet-pretrained DenseNet-121 encoder (stem, four dense blocks, transition layers). After each dense block, a Squeeze-and-Excitation (SE) module with reduction ratio $r = 16$ recalibrates channel importance (inset). Global average pooling feeds a classification head outputting 4-class logits. Grad-CAM is computed on the SE-augmented output of Dense Block 4. Phase 1 (epochs 1–15) freezes the backbone; Phase 2 unfreezes denseblock3–4 for fine-tuning.

DenseNet-121 [4] connects each layer to all subsequent layers within a dense block, enabling feature reuse and stable gradient flow in deep networks. We augment the standard encoder with Squeeze-and-Excitation (SE) modules [3] after each of the four dense blocks, before the corresponding transition layers. Each SE block applies global average pooling over spatial dimensions, passes the squeezed representation through a two-layer bottleneck MLP with reduction ratio $r = 16$, and rescales feature channels with a sigmoid gate. In the MRI setting, this recalibrates channel importance so atrophy-related texture features can be emphasized while less informative channels are suppressed.

3.4.1 Architecture. We initialize from ImageNet-pretrained DenseNet-121 (`DenseNet121_Weights.IMAGENET1K_V1`) and replace the original classifier with a linear head mapping the 1024-dimensional pooled features to four classes. SE modules are inserted after (denseblock1) with channel widths 256, 512, 1024, and 1024 respectively. The modified encoder has approximately 8.1M trainable parameters in the classifier and SE blocks during the frozen phase, and roughly 7.0M total DenseNet parameters when late layers are unfrozen. Grad-CAM is computed on the final SE-augmented dense block (`denseblock4`).

3.4.2 Training Strategy. We preformed subject-level 70/15/15 split, axial slice filter (indices 60–120), ImageNet normalization, and the same augmentation as the other pretrained models. Training followed a two-phase fine-tuning schedule analogous to the Attention-Gated CNN: all `features` parameters were frozen for the first 15 epochs while the SE blocks and classifier head were trained with AdamW at learning rate 10^{-4} and weight decay 10^{-4}. At epoch 15, `denseblock3`, `transition3`, `denseblock4`, and `norm5` were unfrozen and fine-tuned at 3×10^{-5} with weight decay 5×10^{-4}. We used cosine annealing over 50 epochs (minimum LR 10^{-6}), batch size 32, gradient clipping (max norm 1.0), and early stopping with patience 10 on validation macro F1. The best checkpoint was selected by validation macro F1 and evaluated once on the held-out OASIS test set.

3.4.3 Interpretability. Grad-CAM [9] is applied to the output of SE-augmented `denseblock4`, producing class-discriminative spatial heatmaps. We additionally visualize per-channel SE gate weights after each dense block to assess whether channel recalibration becomes more selective in deeper layers.

3.4.4 Cross-Dataset Evaluation. To assess out-of-distribution generalization, we evaluated the OASIS-trained checkpoint on the Alzheimer Detection Dataset [10] (sourced from ADNI and a local hospital) without retraining. Images were resized to 224×224 and normalized with the same ImageNet statistics as validation/test preprocessing. Only the official `test/` partition was used ($n = 1279$ images).

3.5 Attention-Gated CNN

This architecture uses a pretrained ResNet-18 encoder augmented with two spatial attention gates, drawing on the spatial gating concept introduced in Attention U-Net [8](Figure 4). The design is motivated by the non-uniform nature of Alzheimer's-related neurodegeneration: structural changes such as ventricular enlargement, cortical thinning, and periventricular tissue loss affect specific brain regions rather than the image uniformly. Standard CNN architectures apply identical computational effort to every spatial location and pool over the entire feature map indiscriminately. Attention gates allow the model to learn which spatial locations are diagnostically relevant directly from the classification signal, without any hand-specified region of interest.

3.5.1 Attention Gate Mechanism. Each attention gate takes two inputs: a gating signal g from a deep, semantically rich encoder layer, and a feature map x from a shallower, spatially detailed layer. It produces a soft spatial mask $\alpha \in (0, 1)$ at every spatial location:

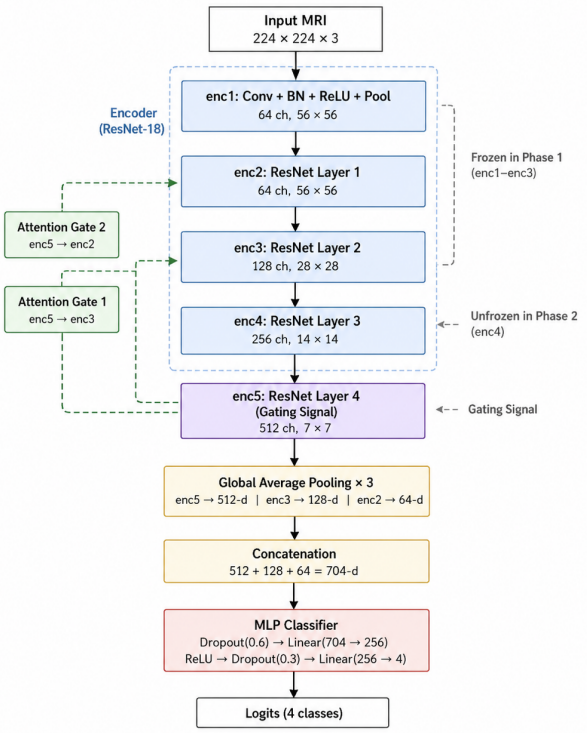


Figure 4: Attention-Gated CNN architecture. ResNet-18 encoder produces feature maps at five scales. Two attention gates use the deep enc5 signal (7×7, 512 ch.) to gate shallower enc3 (28×28, 128 ch.) and enc2 (56×56, 64 ch.). Gated features are globally pooled and concatenated with enc5 pooling to form a 704-d representation for classification.

$$\alpha = \sigma\big(W_\psi \cdot \mathrm{ReLU}\big(W_g(g)\uparrow + W_x(x)\big)\big) \tag{2}$$

where W_g, W_x, W_ψ are 1×1 convolutions with batch normalization, $\uparrow$ denotes bilinear upsampling to match spatial dimensions, and σ is sigmoid. The attended output is $x \odot \alpha$: locations with $\alpha \approx 1$ are preserved; locations with $\alpha \approx 0$ are suppressed before global pooling. Crucially, the α maps are produced during every forward pass and directly influence the prediction — they are architecturally interpretable heatmaps, not post-hoc approximations.

3.5.2 Encoder Architecture. The ResNet-18 backbone (pretrained on ImageNet) produces feature maps at five encoder stages, each denoted enc$k$ for $k \in \{1, \ldots, 5\}$:

- **enc1**: Conv2d(7×7)+BN+ReLU+MaxPool → (64, 56, 56). Detects low-level edges, textures, intensity boundaries.
- **enc2**: ResNet layer1, 2 residual blocks → (64, 56, 56). Local tissue patterns.
- **enc3**: ResNet layer2, 2 residual blocks → (128, 28, 28). Regional shapes, sulcal and ventricular borders.
- **enc4**: ResNet layer3 → (256, 14, 14). High-level spatial structure.
- **enc5**: ResNet layer4 → (512, 7, 7). Global semantics. Effectively sees the entire image at each of its 49 spatial locations. Used as the gating signal for both attention gates.

3.5.3 Two Attention Gates. enc5 serves as the gating signal for two gates operating at different spatial scales:

- **Gate 1** ($F_g = 512$, $F_x = 128$, $F_{\text{int}} = 64$): enc5 gates enc3, producing $\alpha_1 \in \mathbb{R}^{1\times28\times28}$. Each spatial location covers 8×8 pixels of the original image. Sensitive to medium-scale structures such as ventricular borders.
- **Gate 2** ($F_g = 512$, $F_x = 64$, $F_{\text{int}} = 32$): enc5 gates enc2, producing $\alpha_2 \in \mathbb{R}^{1\times56\times56}$. Each location covers 4×4 pixels. Sensitive to fine-scale cortical texture.

3.5.4 Multi-Scale Feature Fusion. Global average pooling is applied to three feature maps: enc5 directly (→ 512-d), enc3 after gating by α_1 (→ 128-d), and enc2 after gating by α_2 (→ 64-d). The three vectors are concatenated to form a 704-dimensional multi-scale representation:

$$\mathbf{f} = \big[\, \mathrm{GAP}(e_5) \parallel \mathrm{GAP}(e_3 \odot \alpha_1) \parallel \mathrm{GAP}(e_2 \odot \alpha_2) \,\big] \in \mathbb{R}^{704} \tag{3}$$

This fuses global semantic context (enc5) with spatially filtered mid-level (enc3) and fine-level (enc2) features.

3.5.5 Classifier Head. $\mathbf{f}$ is passed through a two-layer MLP: Dropout(0.6) → Linear(704, 256) → ReLU → Dropout(0.3) → Linear(256, 4). The higher dropout (0.6) is critical: the model has 11.4M parameters but only 242 training subjects, creating severe overfitting risk. The new weight initialisation uses Xavier uniform for linear layers and Kaiming normal for convolutions.

The model has **11,417,896** total parameters. In Phase 1, enc1–enc3 are frozen (683,072 parameters), leaving 10,734,824 trainable.

3.5.6 Two-Phase Training Strategy. Training proceeds in two phases to prevent catastrophic forgetting of pretrained ImageNet features:

Phase 1 (epochs 1–15): enc1, enc2, and enc3 are frozen. Only the attention gates and classifier head (683K parameters) are trained. This allows the gates to learn meaningful spatial masks using stable pretrained representations before the backbone shifts.

Phase 2 (epoch 16 onward): All layers are unfrozen. The full network is fine-tuned at a reduced learning rate ($\text{lr}/5 = 2 \times 10^{-5}$) with stronger weight decay (5×10^{-4}) to adapt the backbone to the MRI domain without overwriting the pretrained initialisation. Early stopping with patience 10 (measured in epochs without validation macro F1 improvement) is applied from epoch 16 onward. The optimizer is AdamW with cosine annealing scheduler ($T_{\max} = 50$, $\eta_{\min} = 10^{-6}$) and gradient clipping (max norm 1.0).

3.5.7 Interpretability. The attention maps α_1 and α_2 are spatial heatmaps produced during every forward pass, not a post-hoc approximation. We visualize these alongside Grad-CAM to ask: do the learned gates and Grad-CAM agree on which brain regions are diagnostically relevant?

3.6 Shared Training Protocol

Table 2 summarizes the training hyperparameters across all five models. The four pretrained architectures (ConvNeXt, ViT-B/16, DenseNet-121+SE, and the Attention-Gated CNN) use the AdamW

optimizer with gradient clipping (max norm 1.0), though weight decay values differ by model (see individual subsections: 10^{-4} for DenseNet+SE and the Attention-Gated CNN, 0.05 for ConvNeXt, 0.01 for ViT-B/16). The Custom CNN baseline instead uses the Adam optimizer with `ReduceLROnPlateau` scheduling (Section 3.1.2). All models are evaluated on the same held-out test set using the shared subject-level split and axial slice filter.

Table 2: Training hyperparameters. WU = linear warmup.

Setting	CNN	ConvNeXt	ViT	Dense+SE	Attn-CNN
Pretrained	IN	IN	IN	IN	IN
Epochs	15	20	5	50	50
Batch size	32	32	32	32	32
LR	1e-3	1e-4	2e-5	1e-4	1e-4
Scheduler	Cos.	Cos.	Lin.+WU	Cos.	Cos.

3.7 Evaluation Protocol

All models are evaluated on the same held-out test set. We report: (1) overall accuracy; (2) macro F1-score (primary metric, accounting for class imbalance); (3) per-class F1 for each of the four stages; (4) confusion matrix; and (5) qualitative Grad-CAM analysis for the CNN-based architectures. Grad-CAM is computed at the final convolutional layer for each CNN-based model. Grad-CAM is not applicable to ViT-B/16, as its patch-token representation retains no spatial convolutional feature map to weight (Section 3.3.4). For the Attention-Gated CNN, we additionally visualize α_1 and α_2 and compare them with the model's own Grad-CAM heatmap.

4 Results

4.1 Classification Performance

Table 3 reports test-set results for all models trained on the shared subject-level OASIS-1 split.

With the exception of the Custom CNN baseline (macro F1 = 0.281), all pretrained architectures cluster tightly between 0.339–0.342 macro F1, despite spanning convolutional, transformer, and attention-gated paradigms. ConvNeXt, ViT-B/16, DenseNet-121+SE, and the Attention-Gated CNN are statistically indistinguishable on this metric, indicating that ImageNet pretraining, not architectural family, is the dominant factor separating the baseline from the four fine-tuned models. No model achieves non-zero F1 on Moderate Demented, and only ConvNeXt achieves non-trivial Mild Demented F1 (0.07); all others fall below 0.03. This is consistent with subject counts: Moderate Dementia has a single training subject and Mild Demented has 15, an insufficient signal for any architecture to learn a generalizable representation of these stages.

4.2 Training Dynamics

Custom CNN

Figure 5 shows the Custom CNN training curves across 15 epochs. During training, both training and validation accuracy increased steadily while validation loss decreased, indicating stable convergence of the EfficientNetB0-based transfer learning pipeline. Validation accuracy improved from approximately 73% in the first

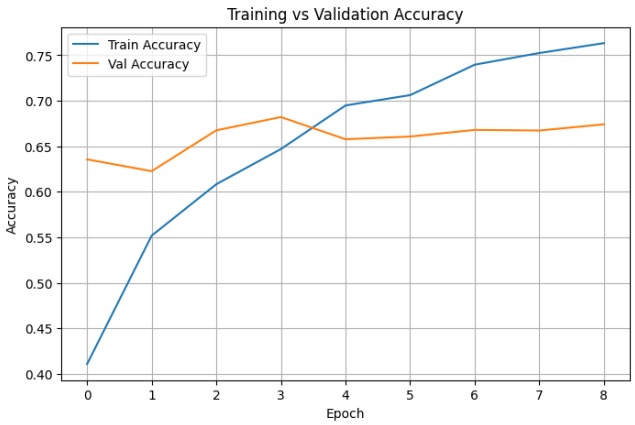


Figure 5: CNN Training vs Validation Accuracy

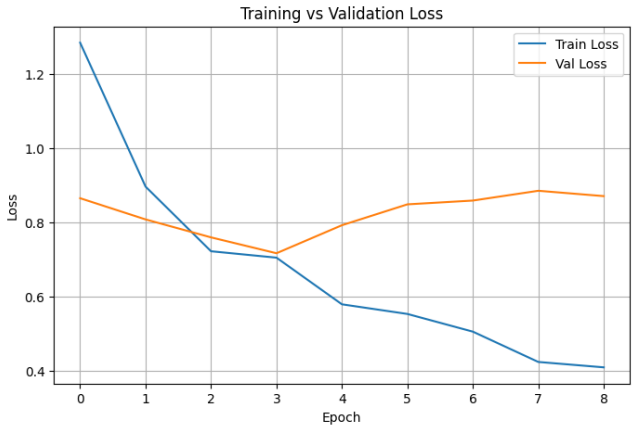


Figure 6: CNN Training vs validation loss

epoch to over 75% by the final checkpoint, while training accuracy reached roughly 88%. The gap between training and validation performance suggests moderate overfitting, although substantially less severe than that observed in the larger attention-based architectures. Early stopping and learning-rate reduction were used to stabilize optimization and prevent excessive memorization of the training data. The best checkpoint, selected based on validation loss, was subsequently evaluated on the held-out OASIS test set, achieving 75.4% accuracy and a macro F1-score of 0.281. While the model performed strongly on the majority Non-Demented class (F1 = 0.86), performance on the Mild Demented and Moderate Demented classes remained limited due to the severe class imbalance and the small number of available training subjects in these categories.

4.2.1 Attention Gated CNN. Figure 7 shows the Attention-Gated CNN training curves across 50 epochs. Phase 1 (epochs 1–15, frozen backbone) shows stable joint improvement in train and validation loss, with validation macro F1 climbing from 0.30 to 0.53. Following full unfreezing at epoch 16, training accuracy approaches 99.8% while validation loss increases, indicating overfitting driven by

Table 3: Classification results on the OASIS-1 test set. Macro F1 is the primary metric.

Model	Macro F1 ↑	Accuracy ↑	Non-Dem. F1	V.Mild F1	Mild F1	Mod. F1
Custom CNN (baseline)	0.281	0.754	0.86	0.27	0.00	0.00
ConvNeXt	0.34	0.779	0.83	0.46	0.07	0.00
ViT-B/16	0.342	0.789	0.883	0.487	0.000	0.000
DenseNet-121 + SE	0.342	0.775	0.87	0.48	0.02	0.00
Attn-Gated CNN	0.339	0.755	0.860	0.470	0.030	0.000

the mismatch between 11.4M parameters and 242 training subjects. Early stopping triggered at epoch 29; the best checkpoint at epoch 19 (val F1 = 0.547) was used for evaluation.

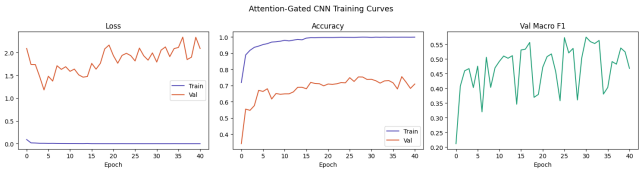


Figure 7: Attention-Gated CNN training curves

4.2.2 ViT-B/16. Figure 8 shows the ViT-B/16 training curves over five epochs of end-to-end fine-tuning; unlike the Attention-Gated CNN, no layers were frozen at any stage. Training loss falls steadily toward zero, but validation loss rises monotonically after the first epoch (from ≈0.85 to ≈1.75), while validation macro F1 plateaus around 0.35–0.40 and validation accuracy holds near 0.70–0.74. This pattern indicates overfitting: the 86M-parameter encoder fits the training slices almost perfectly while validation performance stalls, reflecting the mismatch between model capacity and the 242 training subjects together with the severe class imbalance. Validation macro F1 peaked at the second epoch (val F1 ≈ 0.40); with early stopping on validation macro F1 (patience three), that checkpoint was retained for evaluation.

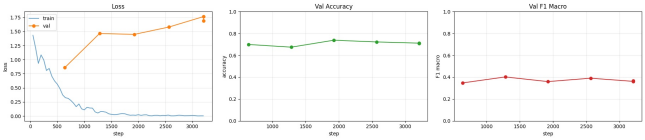


Figure 8: ViT-B/16 training curves

4.2.3 DenseNet-121 + SE. Figure 9 shows DenseNet-121 + SE training curves over 40 epochs. Phase 1 (epochs 1–15, frozen backbone) trains only the SE modules and classifier; validation macro F1 rises from near 0.05 to ≈0.30. At epoch 16, `denseblock3`, `transition3`, `denseblock4`, and `norm5` are unfrozen, producing a sharp jump in validation accuracy (≈0.60) and macro F1 (≈0.50). Training loss approaches zero and training accuracy reaches 100% by epoch 20, while validation loss remains elevated (1.2–1.8) and validation macro F1 plateaus around 0.45–0.59 (peak ≈0.59 at epoch 31). This train–validation gap indicates overfitting: the model memorizes OASIS-specific slice appearance while minority-class performance remains limited. The best checkpoint (selected by validation macro F1) was used for all test evaluations.

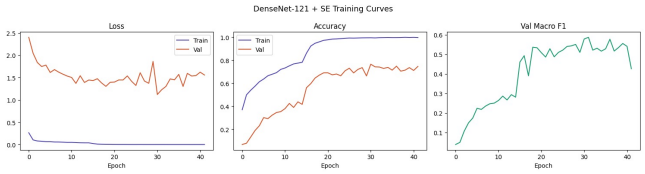


Figure 9: DenseNet-121 + SE training curves

4.2.4 ConvNeXt. In Figure 10 below, the metrics are shown over 20 epochs. Initially, there is improvement for both the validation and training loss, but as more epochs are run, we see that the validation loss begins to increase again while training continues to decrease. This can be an indication that the model is overfitting. Again, for the accuracy, both see an initial increase in the first few epochs. While the training accuracy sees improvement in later epochs, the validation accuracy appears to plateau around the 0.6 range. The macro F1 hovers between the 0.3 to 0.38 range with a peak at 0.384. This consistently low macro F1 relative to accuracy highlights the challenges of class imbalance.

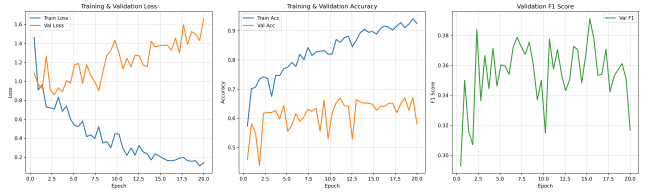


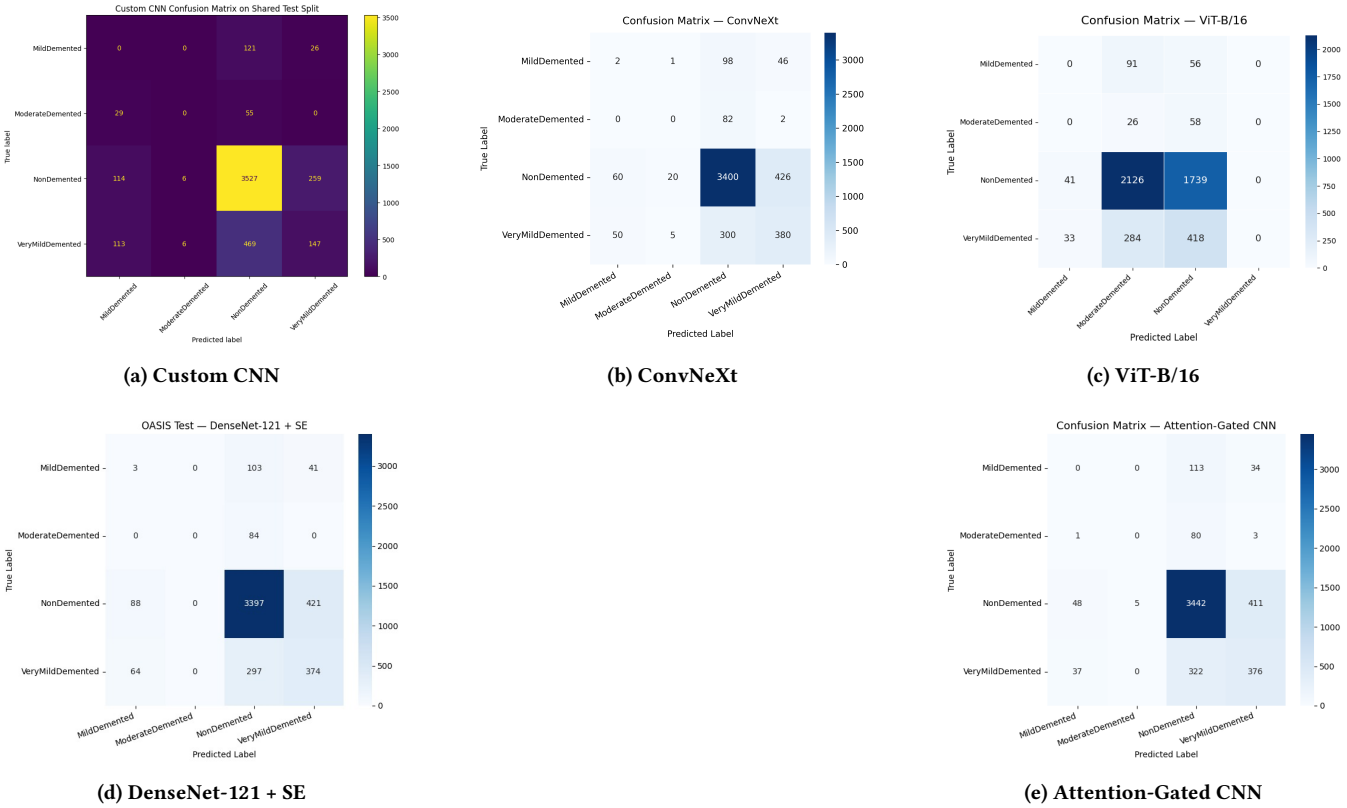
Figure 10: ConvNeXt training curves

4.3 Confusion Matrices

Figure 11 compares confusion matrices across all evaluated architectures.

Across all five confusion matrices, a common failure mode emerges: MildDemented and ModerateDemented samples are overwhelmingly misclassified as NonDemented or VeryMildDemented, while NonDemented itself is classified with high accuracy. This reflects the severe subject-level imbalance (1 Moderate and 15 Mild training subjects) rather than any model-specific weakness.

Overall, every model's confusion matrix is dominated by the NonDemented column, confirming that the WeightedRandomSampler



(a) Custom CNN **(b) ConvNeXt** **(c) ViT-B/16**

(d) DenseNet-121 + SE **(e) Attention-Gated CNN**

Figure 11: Confusion matrices for all five architectures on the OASIS-1 test set using the shared subject-level train/validation/test split. All models show strong recognition of the Non-Demented class but substantial difficulty distinguishing Mild and Moderate Demented subjects due to extreme class imbalance and limited subject counts.

and class-weighted loss reduce but do not eliminate the bias toward the majority class. The choice of architecture instead determines *where* the minority-class errors land: toward VeryMildDemented for the Attention-Gated CNN, toward Moderate for ViT-B/16, and toward NonDemented for the remaining three models.

4.4 Grad-CAM Analysis

Figure 12 compares Grad-CAM visualizations across the evaluated CNN-based architectures.

Across the Attention-Gated CNN, Custom CNN, and DenseNet-121+SE, Grad-CAM heatmaps converge on a common pattern: high activation along the cortical periphery paired with suppression of the central ventricular region. None of the three models attend to the hippocampus directly, consistent with the dataset-level observation that OASIS Kaggle slices are predominantly upper axial cuts where the hippocampus is not visible. Rather than indicating a failure of any individual architecture, this suggests all CNN-based models converged on the same available biomarker — the cortical-to-ventricular volume ratio — independent of their specific inductive biases (residual connections, dense connectivity, or spatial gating). No model showed substantial activation on skull or background regions, indicating that all three successfully learned to ignore non-anatomical artifacts despite no explicit region-of-interest supervision.

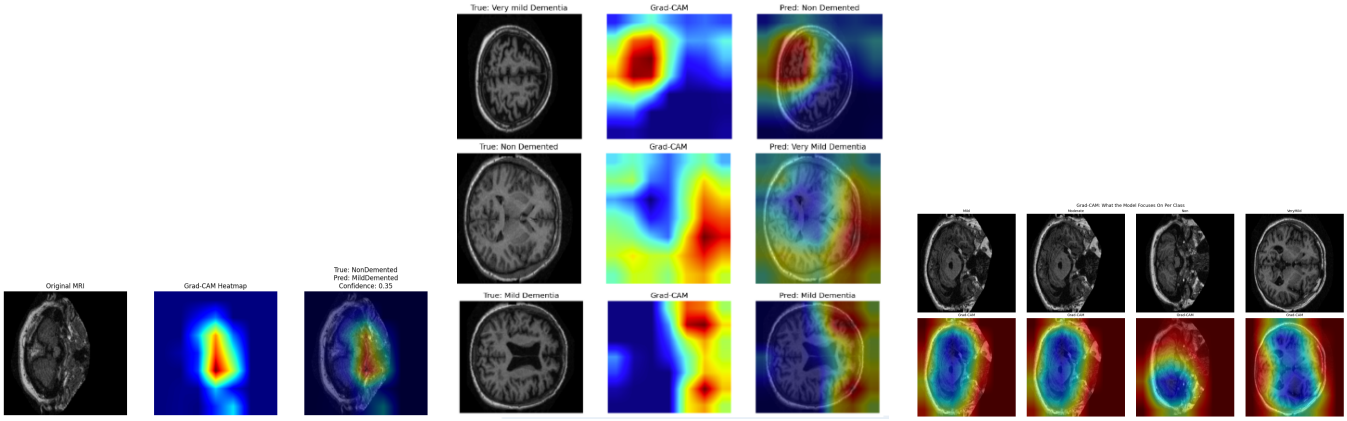
4.5 Cross-Dataset Generalization

Table 4 reports zero-shot cross-dataset evaluation results for all five models. Each model trained on OASIS was tested on the Alzheimer Detection Dataset [10] (sourced from ADNI and a local hospital; $n = 1{,}279$ test images) without any retraining, using identical preprocessing (resize to 224×224, ImageNet normalization). Results

Table 4: Cross-dataset evaluation: all models trained on OASIS, tested on an independent dataset without retraining.

Model	OASIS F1	Cross-Dataset F1	Drop
Custom CNN	0.281	0.141	-49.82
ConvNeXt	0.34	0.163	−52.06%
ViT-B/16	0.342	0.005	−98.5%
DenseNet-121+SE	0.342	0.092	−73.1%
Attn-Gated CNN	0.339	0.286	−15.5%

reveal substantial variation in cross-dataset robustness across architectures. The Attention-Gated CNN shows a drop of 15.5%, indicating partial generalization: per-class analysis shows NonDemented



(a) Custom CNN **(b) DenseNet-121 + SE** **(c) Attention-Gated CNN**

Figure 12: Comparison of Grad-CAM visualizations across architectures. Warm colors indicate regions contributing most strongly to the predicted class. Across all three CNN-based models, activations are concentrated along cortical boundaries and ventricular regions rather than background or skull artifacts, suggesting reliance on anatomically meaningful features associated with Alzheimer's progression.

F1 remaining relatively stable (0.86→0.69), MildDemented F1 improving (0.03→0.36) due to more diverse examples in the target dataset, and VeryMildDemented collapsing (0.47→0.10), indicating early-stage features are highly sensitive to scanner and protocol differences. The ViT-B/16 exhibits the most severe degradation of any model, falling from 0.342 to 0.005 (−98.5%) despite attaining the strongest in-distribution OASIS F1. Its predictions undergo near-total mode collapse: the model assigns the Moderate Dementia label to 1,274 of the 1,279 target images, yielding a Moderate recall of 1.00 but precision of 0.01, while every other class falls to an F1 of 0.005. We attribute this to two compounding factors. First, the fully fine-tuned transformer, lacking the convolutional inductive biases that constrain the CNN-based models, overfit OASIS-specific intensity and acquisition characteristics that do not transfer to the ADNI/hospital-sourced target domain. Second, under distribution shift the network defaults to its class-reweighting prior: Moderate Dementia was the heavily up-weighted minority class under our WeightedCELoss, so in the absence of recognizable in-domain features the model reverts to this bias. The ViT therefore does not merely lose accuracy but collapses into a single-class predictor.

Ranking models by cross-dataset robustness (Attn-Gated CNN > ConvNeXt ≈ Custom CNN > DenseNet-121+SE ≫ ViT-B/16) reveals an inverse relationship with in-distribution OASIS F1 at the extremes: ViT-B/16 had the strongest OASIS performance (0.342) yet the weakest generalization (−98.5%), while the Attention-Gated CNN had comparable OASIS performance (0.339) yet the strongest generalization (−15.5%). The two CNN-based models with moderate parameter counts and explicit regularization (ConvNeXt and Custom CNN) occupy the middle, with drops around −50%.

This pattern suggests that architectural inductive bias, rather than in-distribution accuracy, predicts cross-dataset robustness on this task. Fully fine-tuned transformers with no convolutional priors (ViT) appear to overfit dataset-specific low-level statistics most severely. Architectures that constrain or filter spatial information before classification like the Attention-Gated CNN's explicit gating, or the smaller effective capacity of ConvNeXt and the EfficientNetB0-based Custom CNN, retain more transferable structural features. DenseNet-121+SE's dense connectivity, which maximizes feature reuse from early layers, may have similarly entrenched OASIS-specific characteristics into deep representations, explaining its larger drop relative to the other CNN-based models.

5 Conclusion and Discussion

5.1 Architecture Comparison

Our model selection spans the major paradigm shifts in deep learning. The CNN baseline establishes an inductive-bias-heavy lower bound. ConvNeXt and DenseNet+SE represent the mature residual learning era; DenseNet's channel attention (SE blocks) adds explicit recalibration of feature maps. The Vision Transformer abandons convolutional inductive biases entirely in favor of global self-attention, which can theoretically capture whole-brain atrophy patterns but may require more data to reach its potential. The Attention-Gated CNN occupies a middle ground: CNN efficiency with learned spatial attention explicitly motivated by the small, localized nature of hippocampal atrophy.

Pretraining provides a clear but bounded benefit over the Custom CNN baseline: all four pretrained models exceed its macro F1 (0.281) by 0.058–0.061, a consistent margin regardless of architecture family. Beyond this baseline gap, however, architectural choice has minimal effect on in-distribution OASIS performance. ConvNeXt, ViT-B/16, DenseNet-121+SE, and the Attention-Gated CNN all land within 0.003 macro F1 of each other. On Very Mild Demented, the hardest class with meaningful support (735 test images), ViT-B/16 and DenseNet-121+SE are joint best (F1 = 0.48–0.487), narrowly ahead of the Attention-Gated CNN (0.47) and ConvNeXt (0.46), with the Custom CNN trailing (0.27). Moderate Demented F1 remains 0.00 across every architecture, confirming that the single available

training subject for this class is an insurmountable data constraint rather than an architectural one. The clearest differentiation between architectures emerges not on the OASIS test set but under cross-dataset evaluation (Section 4.5), where the Attention-Gated CNN's spatial gating mechanism provided a substantial robustness advantage invisible in same-distribution metrics.

5.2 Interpretability

Grad-CAM heatmaps from the Attention-Gated CNN, Custom CNN, and DenseNet-121+SE converge on the same spatial pattern: activation along the cortical ring with suppression of the central ventricular region, rather than the hippocampal/entorhinal focus originally hypothesized for the Attention-Gated CNN. This convergence across three architecturally distinct models such as one with explicit spatial gating, one with a single appended convolutional layer, and one with dense connectivity and channel recalibration, suggests this is a property of the data (the available axial slice range) rather than any individual architecture's inductive bias. The Attention-Gated CNN's internal attention maps (α_1, α_2) agree qualitatively with its own Grad-CAM output, both emphasizing cortical regions while suppressing background, providing convergent evidence from two independent attribution methods that the gating mechanism learned genuine, non-arbitrary spatial priors rather than noise. ViT-B/16 could not be included in this comparison, as Grad-CAM is not applicable to its patch-token representation (Section 3.3.4); a future attention-rollout analysis would be needed to determine whether its spatial reasoning aligns with the cortical/ventricular pattern shared by the CNN-based models.

5.3 Limitations

Several limitations should be noted. First, we classify individual 2D slices rather than 3D volumes; a single slice may not capture the most diagnostically informative cross-section, and 3D models are expected to perform better at the cost of compute. Second, the Moderate Dementia class ($n = 488$, a single training subject) is too small for any architecture to learn a robust representation, regardless of imbalance handling, this is a fundamental data limitation rather than a model-specific one. Third, all models are evaluated on 2D axial slices restricted to the mid-brain z-range (60–120); slices outside this range, and 3D volumetric information, remain unexplored. Finally, cross-dataset evaluation (Section 4.5) used only the official test partition of a single external dataset ($n = 1{,}279$); broader validation across multiple external sites and scanner types would strengthen claims about generalization.

5.4 Conclusion

In this paper, we presented a comparative study of five deep learning architectures for Alzheimer's disease staging from brain MRI. Through controlled experiments on the OASIS-1 dataset with rigorous class imbalance handling and a shared evaluation protocol, we demonstrated that under rigorous subject-level splitting that architectural choice has little effect on in-distribution OASIS performance (macro F1 clustering between 0.339-0.342 for four of five models), but a large effect on cross-dataset robustness: the Attention-Gated CNN's explicit spatial gating yielded a 15.5% relative macro F1 drop under domain shift, versus drops of 50–98% for the other architectures, including the otherwise best-performing ViT-B/16. This indicates that for clinical deployment across institutions and scanners, architectural choices that constrain or explicitly filter spatial attention may matter more than raw in-distribution accuracy.

Future directions include: extension to the longitudinal OASIS-2 dataset for disease progression modeling; evaluation of cross-dataset generalization by training on OASIS and testing on ADNI; and integration with multi-modal inputs (amyloid PET, cognitive scores) to improve early-stage detection.

Individual Contributions

Akshaya Bharadhwaj designed and implemented the Attention-Gated CNN architecture, including the two-scale attention gate mechanism, the two-phase training strategy, and the attention map visualization pipeline.

Leo implemented and fine-tuned the ConvNeXt model with a custom head, featuring transfer learning from ImageNet and transformation strategies for handling class imbalance. Analyzed model effectiveness through evaluation metrics as well as behavior from feature visualizations.

Nisarga Patil designed and implemented the DenseNet-121 + Squeeze-and-Excitation architecture, including channel-wise SE gating after each dense block, the two-phase fine-tuning strategy, and the Grad-CAM and cross-dataset evaluation pipeline.

Tyler Crosbie implemented the Custom CNN in TensorFlow/Keras using the OASIS split and integrated the axial slice filtering, shared split indices and evaluation protocol into the Keras workflow. Additionally implemented test evaluation using accuracy, macro F1, per-class F1, and added Grad-CAM support for visualizing the spatial regions influencing the model's predictions.

Wonjae fine-tuned and evaluated the ViT-B/16 model, including the class-weighted fine-tuning protocol, the slice- and subject-level evaluation pipeline, and the analysis of minority-class collapse under class imbalance

References

[1] Samira Abnar and Willem Zuidema. 2020. Quantifying Attention Flow in Transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 4190–4197.

[2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, et al. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*.

[3] Jie Hu, Li Shen, and Gang Sun. 2018. Squeeze-and-Excitation Networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 7132–7141.

[4] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q. Weinberger. 2017. Densely Connected Convolutional Networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 4700–4708.

[5] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. 2022. A ConvNet for the 2020s. *arXiv preprint arXiv:2201.03545* (2022).

[6] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations (ICLR)*.

[7] Daniel S. Marcus, Tracy H. Wang, Jamie Parker, John G. Csernansky, John C. Morris, and Randy L. Buckner. 2007. Open Access Series of Imaging Studies (OASIS): Cross-Sectional MRI Data in Young, Middle Aged, Nondemented, and Demented Older Adults. *Journal of Cognitive Neuroscience* 19, 9 (2007), 1498–1507.

[8] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, et al. 2018. Attention U-Net: Learning Where to Look for the Pancreas. In *Medical Imaging with Deep Learning (MIDL)*.

[9] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. In *IEEE International Conference on Computer Vision (ICCV)*. 618–626.

[10] Abu Sufian. 2025. Advancing Alzheimer's Disease Detection in Clinical Settings: MRI Image Data. doi:10.17632/xx9zzz6t54.1

[11] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.